

MDD10A Motor Driver

The MDD10A motor driver is designed to control two brushed DC motors independently using a full H-Bridge configuration based on efficient discrete MOSFETs.

It supports a motor supply voltage range of 5V to 30V DC, accommodating various power sources including common battery configurations (6V, 12V, and 24V) as well as regulated power supplies. This flexibility makes it suitable for applications ranging from small robots to industrial-grade motor systems.

The driver is engineered to handle continuous currents of up to 10A per channel without needing external heatsinks or fans, thanks to its optimized PCB layout with thick copper traces and high-efficiency MOSFETs. For short-duration current surges, such as during motor startup or when encountering obstacles, it can tolerate peak currents up to 30A for several milliseconds without damage. This robust design ensures reliability under dynamic load conditions.

A key design feature of the MDD10A is its separate power inputs for motor (VM) and logic (VCC) circuits. VM powers the motors directly, while VCC supplies the logic section, usually at 3.3V or 5V.

This allows flexible interfacing with various microcontrollers and embedded systems. Additionally, the onboard 5V linear regulator (200mA max) can be used to power external modules or logic circuits if VM is at least 6V, simplifying the system's power architecture.



Key Ratings :

- Motor Voltage Input (VM): 5V to 30V DC

- Logic Voltage Input (VCC): 3.3V or 5V DC
- Continuous Current per Channel: 10A
- Peak Current per Channel: 30A (short bursts)
- Onboard 5V Regulator Output: 200mA max

Control Interfaces: PWM and DIR

The MDD10A utilizes a simple yet effective control scheme through two pins per motor channel: PWM (Pulse Width Modulation) and DIR (Direction). This interface provides precise control over motor speed and direction without complex communication protocols.

The PWM pin allows for speed control by adjusting the duty cycle of the input signal. The driver supports PWM frequencies up to 20kHz, ensuring silent motor operation—important for noise-sensitive environments. Most modern microcontrollers, including Arduino, Raspberry Pi, and STM32, can easily generate PWM within this frequency range.

The DIR pin sets the direction of rotation. A logic LOW (0) rotates the motor in one direction, while a logic HIGH (1) reverses it. Both PWM and DIR pins support 3.3V and 5V logic levels, ensuring compatibility across different microcontroller platforms.

The motor can be stopped by setting the PWM duty cycle to 0%, placing it in coast mode. Holding DIR constant while varying PWM adjusts speed in a fixed direction; toggling DIR reverses the direction at any speed.

Control Interface Summary :

- Control Signals: PWM (Speed), DIR (Direction)
- PWM Frequency Range: Up to 20kHz
- Logic Level Compatibility: 3.3V and 5V
- Speed Control: Via PWM duty cycle (0–100%)
- Direction Control: DIR LOW = Forward, DIR HIGH = Reverse

Application Circuits and Usage Examples

The MDD10A is suitable for a wide range of applications, from simple DIY projects to advanced robotics and automation systems. Its straightforward pinout and compatibility with standard microcontroller logic make it easy to integrate into various systems.

A basic application involves connecting the motor supply (VM) to a battery or DC power supply between 5V and 30V, with GND shared between the motor driver and control

board. The PWM and DIR pins connect to the microcontroller's GPIOs (preferably PWM-capable pins).

For instance, using an Arduino:

- Speed control: `analogWrite(PWM_PIN, value)` where value ranges from 0 (stop) to 255 (full speed)
- Direction control: `digitalWrite(DIR_PIN, HIGH/LOW)`

In robotic applications, such as differential drive systems, each motor (left/right wheel) can be independently controlled for smooth turning and manoeuvrability. The MDD10A's built-in protections (overcurrent, thermal shutdown, undervoltage lockout) safeguard the system during unexpected conditions, enhancing operational safety and longevity.

The driver's passive cooling design allows it to operate at full rated current without active cooling; however, external airflow may be used in high-demand environments for thermal stability.

Application Highlights (Bulleted Summary):

- Ideal for: Robotics, RC Vehicles, AGVs, DIY Automation
- Microcontroller Compatibility: Arduino, Raspberry Pi, STM32, ESP32
- Built-in Protections: Overcurrent, Thermal Shutdown, Undervoltage Lockout
- Thermal Design: No heatsink/fan required for typical use